

WALMART SALES PERFORMANCE DASHBOARD

Using Microsoft Power BI Project Report

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Tool Used: Microsoft Power BI Desktop

Introduction

For this project, I analysed Walmart's sales data using Microsoft Power BI. The main aim was to understand the sales performance of different stores and identify useful patterns in the data.

The dashboard helps analyse weekly sales, store performance, holiday and non-holiday sales, sales trends over time and the relationship between fuel price and sales. I also created a business insights page to present key findings and recommendations.

Objectives

The main objectives of this project were:

- To analyse Walmart's weekly sales performance.
- To compare sales across different stores.
- To identify sales trends over time.
- To compare holiday and non-holiday sales.
- To analyse the relationship between fuel price and weekly sales.
- To create an interactive dashboard using Power BI.
- To identify business insights and provide recommendations.

Dataset Description

The dataset contains **6,436 records from 45 Walmart stores**. It includes sales data along with other factors such as temperature, fuel price, CPI and unemployment.

The main fields used were:

- **Store** – Identifies the Walmart store.
- **Date** – Date of the sales record.
- **Weekly Sales** – Total sales recorded for a store in a week.
- **Holiday Flag** – Shows whether the week was a holiday or non-holiday.

- **Temperature** – Temperature during the period.
- **Fuel Price** – Fuel price during the week.
- **CPI** – Consumer Price Index.
- **Unemployment** – Unemployment rate.

These fields helped in analysing sales performance from different business perspectives.

Data Preparation

The Walmart dataset was imported into Power BI for analysis. The available fields were checked and used according to their respective data types.

The dataset was organised around sales-related fields such as **Store, Date, Weekly Sales, Holiday Flag, Fuel Price, CPI, Temperature and Unemployment**. The prepared data was then used to create measures and visualizations for the dashboard.

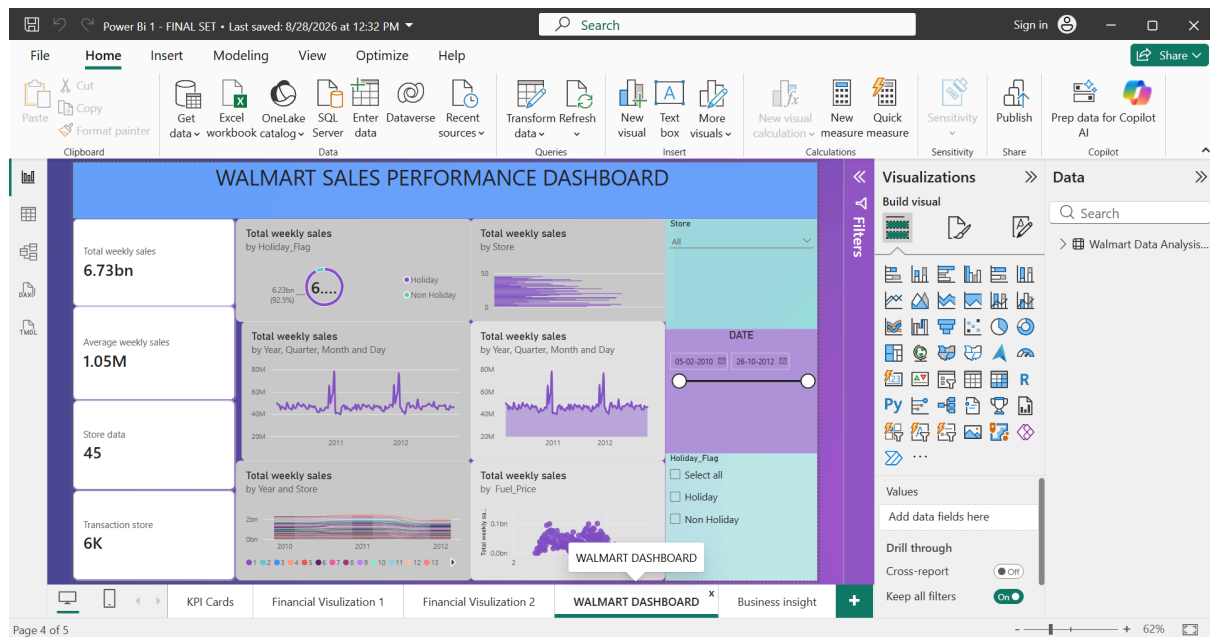
DAX Measures Used

I created measures in Power BI to calculate and display important sales information through KPI cards.

The main measures used were:

- **Total Weekly Sales**
- **Average Weekly Sales**
- **Store Data**
- **Transaction Store**

These measures helped present Walmart's overall performance in a simple, easy-to-understand format.



Dashboard Components

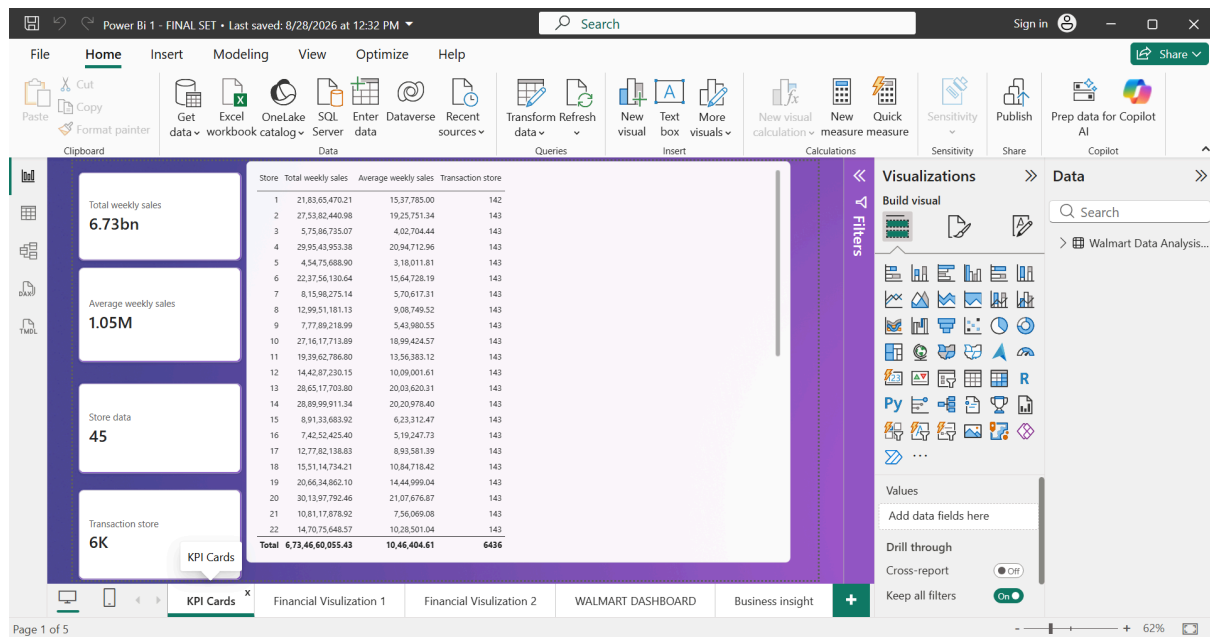
KPI Cards

The KPI section provides a quick overview of the dataset and sales performance.

The dashboard displays:

- **Total Weekly Sales – 6.73 billion**
- **Average Weekly Sales – 1.05 million**
- **Number of Stores – 45**
- **Transactions per Store – approximately 6K**

These KPIs make it easier to understand the overall performance at a glance.



Sales by Holiday Flag

A donut chart was used to compare sales during **holiday and non-holiday periods**.

The dashboard shows approximately **7.5% of sales from holiday periods and 92.5% from non-holiday periods**. This helps understand how much of the overall sales comes from holiday periods.

Total Weekly Sales by Store

A bar chart was used to compare the total weekly sales of different Walmart stores.

This visualization makes it easier to identify **higher-performing and lower-performing stores**. From the analysis, **Store 20 recorded the highest sales performance**.

Weekly Sales Trend

A line/area chart was used to show how weekly sales changed over time.

The chart shows fluctuations in sales and noticeable increases during certain periods. This helps in understanding changes in Walmart's sales performance across different dates.

Sales by Year and Store

A visualization comparing **year and store** was created to understand how the performance of different stores changed over time.

This helps identify stores that maintained strong performance, as well as those whose sales changed across years.

Fuel Price vs Weekly Sales

A scatter chart was used to study the relationship between **fuel price and weekly sales**.

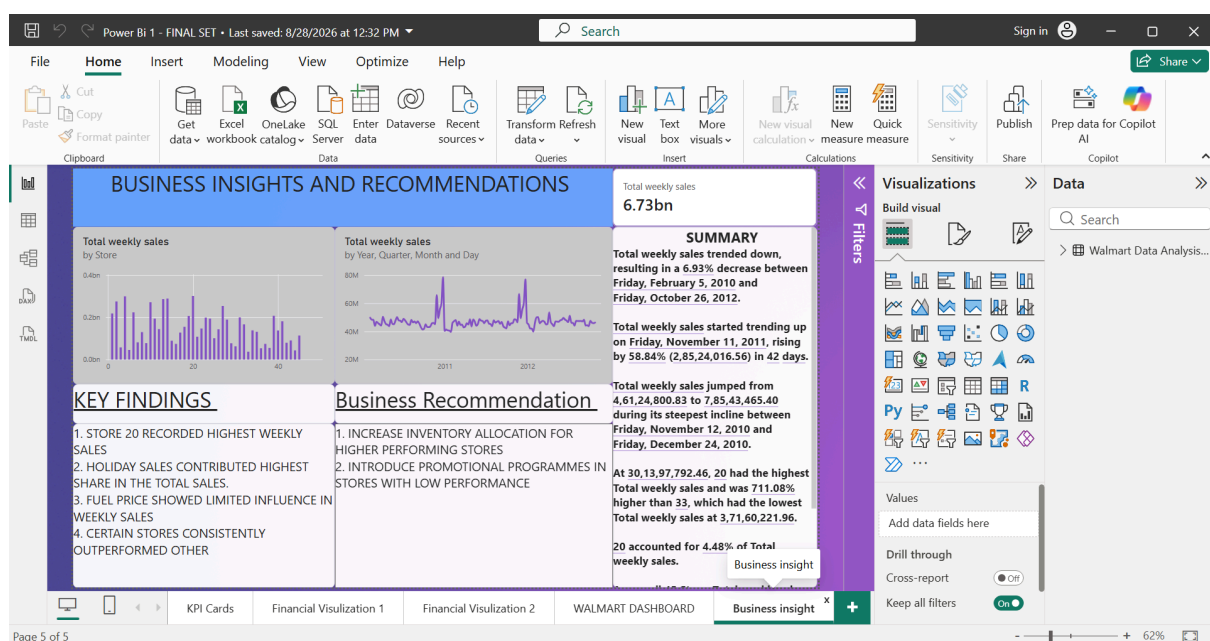
The points are widely distributed and do not show a strong clear relationship. Based on the dashboard analysis, **fuel price appears to have limited influence on weekly sales**.

Interactive Filters

To make the dashboard interactive, the following filters were added:

- **Store**
- **Holiday Flag**
- **Date**

These filters allow users to select specific stores, periods or holiday conditions and analyse the sales data accordingly.



Key Findings

The main findings from the analysis were:

1. **Store 20 recorded the highest weekly sales.**
2. **Holiday sales contributed around 7.5% of total sales.**
3. **Certain stores consistently performed better than others.**
4. **Weekly sales showed noticeable fluctuations over time.**
5. **Fuel price showed limited influence on weekly sales.**
6. The overall sales recorded in the dashboard were approximately **6.73 billion**.
7. The average weekly sales were approximately **1.05 million**.

Business Insights

The analysis provides some useful insights for Walmart's business performance.

Store Performance:

The difference in sales between stores shows that some locations perform much better than others. These stores can be studied to understand the factors contributing to their stronger performance.

Holiday Sales:

Although holiday sales form a smaller share of total sales, holiday periods can still be important for promotional and inventory planning.

Fuel Price:

The scatter chart does not show a strong relationship between fuel price and weekly sales. Therefore, other factors may have a greater influence on sales performance.

Sales Trends:

The changes in weekly sales over time indicate that sales are not constant and can increase or decrease during specific periods.

Business Recommendations

Based on the findings, the following recommendations can be made:

1. **Increase inventory allocation for higher-performing stores** to avoid stock shortages during periods of high demand.
2. **Introduce promotional programmes for lower-performing stores** to improve customer engagement and sales.

3. **Study the practices of high-performing stores** and identify strategies that can be applied to other stores.
4. **Plan inventory and promotions around important sales periods** by using historical sales trends.
5. **Monitor multiple business factors instead of focusing only on fuel prices**, since the analysis does not show a strong relationship between fuel price and sales.

Conclusion

This project helped me understand how Microsoft Power BI can be used to convert sales data into meaningful business information. The dashboard made it easier to compare stores, analyse sales trends, understand holiday sales and examine external factors such as fuel prices.

The interactive features also made the analysis easier to explore. Overall, the project helped me improve my understanding of **data visualization, business analysis and using Power BI to support business decision-making**.